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book 1



THE SCIENCE OF space

WENDY ST GERMAIN & PETER GRIBBEN

CLASSMASTERS

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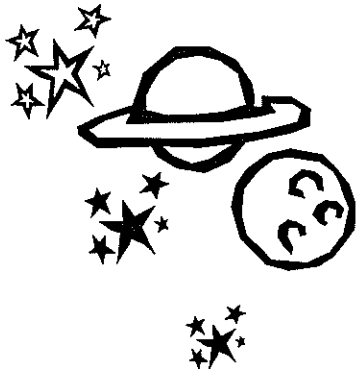
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The Universe: an introduction



Most people who have looked at the skies at night have wondered what it would be like to travel through space.

Over five thousand years ago, in the temples of Memphis, Egypt, the accuracy of the calendars was maintained using the stars. The arrival or departure of familiar stars predicted important annual events, such as the flooding of land near the river Nile.

About three thousand to five thousand years ago, Stonehenge was built in England. This structure of large stones (megaliths) is thought to have been used as a calendar to predict eclipses, equinoxes and solstices.

The entire celestial world is called the *universe*. The universe contains every planet, moon, star and black hole, right down to the tiniest piece of 'space dust'. An *eclipse* occurs when one celestial body is obscured, or overshadowed, by another planet or moon. When day and night are of equal length an *equinox* occurs. The longest and shortest days of each year are called the *summer solstice* and the *winter solstice* respectively. 'Solstice' means 'Sun stands still' which is the way it appears at these times.

1 Define 'eclipse'. _____

2 Define 'equinox'. _____

3 Define 'solstice'. _____

'Solar' means 'of the Sun' and a 'system' is an organised group of things, so the Sun and everything in its vicinity is called the *Solar System*. The Sun is found at the centre of our Solar System with everything else, including nine planets and many moons, travelling around it. Our Sun is fairly average both in age and luminosity (brightness). It is the source of all our energy on Earth. Moons are often called *satellites*, because a satellite is an object that orbits (circles) a larger celestial object.

It is generally believed that our Solar System is about 4500 million years old (debate continues) and is estimated to be about half way through its life. Our Solar System is the only one known to exist, but some scientists think they have found other planets orbiting far away with suns of their own.

4 Name two groups of people who might be concerned about floods from rivers.

_____

5 What is a *megalith*? _____

6 What does 'Solar' mean? _____

7 In which country is the river Nile? _____

8 How many planets are in our Solar System? _____

9 Which celestial object is at the centre of our Solar System? _____

10 What is the source of all energy on Earth? _____

11 What is meant by 'luminosity'? _____

12 Why can we describe the Earth's moon as a *satellite*? _____

_____

13 What is the estimated age of the Solar System? Write the answer in figures and in words.

_____

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Test your Astronomy

VOCABULARY

1 Match the following words to the astronomical descriptions below.

celestial galaxy universe Solar System orbit light year planets

a Described as 'wanderers' by ancient astronomers. _____

b An 'island' of planets, stars, etc. found in the universe. _____

c The distance light travels in one year is called this. _____

d Referring to the sky or 'the heavens'. _____

e Everything that exists in the skies (that is, entire celestial creation). _____

f To travel around an object, following a certain path. _____

g The Sun and all the planets that revolve around it. _____

THE SOLAR SYSTEM

Our Solar System consists of one star and nine planets. The planets are classified into two groups: '*gas giants*' and the '*terrestrial*' planets. Heavy planets are made of rocks and metals. Light planets are made of gases.

2 Name the planets in their correct order, starting from closest to the Sun.

a _____

b _____

c _____

d _____

e _____

f _____

g _____

h _____

i _____

3 State whether each of these planets is a *gas giant* or a *terrestrial* planet:

a Earth _____

b Jupiter _____

c Mars _____

d Mercury _____

e Neptune _____

f Pluto _____

g Saturn _____

h Uranus _____

i Venus _____



4 Identify each of these planets from the details supplied.

a Has four large satellites that were the first moons (apart from Earth's moon) to be discovered using a telescope. _____

b Looks yellow from Earth. Has rings. _____

c Has moons called Phobos and Deimos. _____

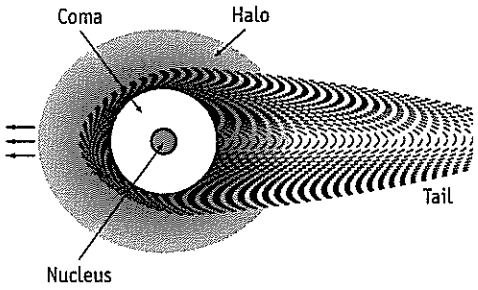
d Has the greatest abundance of heavy elements and the same name as a liquid metal.

_____

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Comets, Asteroids and Meteors

Comets and asteroids are objects that are often sighted first by amateur astronomers. This is because they are easily seen with the naked eye or with a reasonable pair of binoculars.



COMETS

Comets are 'space ice balls' made up of a head and a tail. The tail always faces away from the Sun and can be millions of kilometres long. Comets are usually quite large and are made up of chunks of ice and dirt. They hurtle through space, and though we often miss seeing them, they are not rare.

One famous comet is named Halley's Comet. It was first recorded in 1301. Edmund Halley noted that a certain comet had been reported in 1531, 1607 and 1682. He believed it was always the same comet, predicting it would reappear in 1758. When it did, the comet was named after him. Halley's comet passes Earth once every 76 years. In 1997, the Hale-Bopp comet crossed our skies giving us a marvellous astronomical display. Comets have elliptical orbits that can take from just over three years to thousands of years to complete.

1 Halley's comet was last seen in 1986. When do we expect it again? _____

2 Draw and label three comets orbiting the Sun. Be sure to angle them correctly!

3 The word 'comet' comes from the Latin, 'long haired'. Why is this an apt description? _____

4 Meteor showers often occur in conjunction with the passage of comets. What is a meteor?

_____

5 Compare the orbits of comets with those of the major planets in the Solar System.

Research

6 Find out what materials make up a comet's tail. _____

ASTEROIDS

Asteroids are small planets, the largest being Ceres, with a diameter of 1000 km. Most asteroids occur between the orbits of Mars and Jupiter. It was once believed that the asteroids were remnants of a former planet. One current theory states that most of the 3000 or so asteroids in existence in the Solar System are the result of matter coming together but being unable to form a planet due to the gravitational influence of Jupiter.

7 Asteroids have been described as minor planets. Explain why they can be classed as planets. _____

_____

8 Draw a diagram showing the order of the planets in relation to the Sun. Indicate in this diagram the position of the asteroid 'belt'.

9 'The word "asteroid" comes from the Greek meaning *star-like*'. True or false? _____

METEORS

Meteors are small, solid bodies that enter Earth's atmosphere from outer space. They travel so quickly through the air that the friction between the air and the meteor causes the meteor to heat up and emit light. 'Shooting (falling) stars' are meteors. Most meteors are burnt up as they pass through the atmosphere. A *meteorite* is a meteor that reaches the Earth's surface.

10 Meteors are often described as being *incandescent*. What does 'incandescent' mean? _____

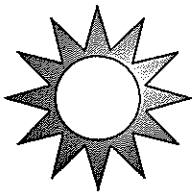
11 Explain the difference between a *meteor* and a *meteorite*. _____

Research

12 Dinosaurs died out quite suddenly millions of years ago. Theories have been put forward that this extinction was caused by sudden climatic change caused by the Earth being hit by a large asteroid. Find out when the dinosaurs died out. Give details of how an asteroid could have caused such a catastrophe.

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The Sun: a massive nuclear reaction I



1 Complete the passage below using words from this list:

second	star	heat	blind	hot	orbit
centre	hydrogen	light	Earth	billions	

All stars are **a** _____ gaseous bodies. The Sun is a **b** _____ we all know. As stars go, it is an average, stable, middle-aged star and one of an estimated 200 million stars in the Milky Way. It is a G2 type star. The Sun is found at the **c** _____ of our Solar System. It is not a solid structure, but an enormous ball of gas where **d** _____ is reacting at a rate of 400 million tons per **e** _____. The product of this fusion reaction is helium and heat. Among stars, the sun is not outstanding, yet life on **f** _____ could not survive without its **g** _____ and **h**_____. Scientists estimate that the Sun has been giving out energy at around its present rate for **i** _____ of years. The temperature at the Sun's core is estimated to be 15 million°C and its surface temperature is a cool 6000°C. In the same way hot fat splatters out of a frying pan, so do burning tongues of fire 'splatter' off the Sun. The Sun also orbits the galaxy, taking about 250 million years to complete one **j** _____. This is called a *cosmic year*. The Sun is so bright it will **k** _____ you if you stare at it for any length of time. So remember, if you do not wish to risk damaging the retina of your eyes, never look directly at the Sun.

SUNSPOTS

Using special equipment, scientists have discovered *sunspots*. Sunspots are dark regions on the Sun's surface where the temperature is lower. Chinese astronomers may have observed sunspots over 2000 years ago. Galileo used a telescope early in the seventeenth century to observe and record sunspots. Sunspots seem to follow a cycle of 11 (Earth) years. Often, when sunspots are created, spectacular light shows, known as *Aurorae*, that can be seen at the Earth's poles, occur.

SOLAR WINDS

As the Sun 'burns', great tongues of fire are sent out from its surface. From these fiery flares tiny, charged particles stream out and can react with gases in the lower atmosphere. Such reactions are also thought to contribute to *Aurorae*. The particles can also become trapped in the Earth's magnetic field, creating electrically charged belts around the Earth's equator. Solar flares can cause disruption to radio communication on Earth.

2 Despite the Sun using millions of tonnes of hydrogen a second, why do scientist estimate it can continue to give out its current rate of energy for billions of years?

_____

_____

3 What are 'sunspots'? _____

_____

4 Why could it be dangerous to study sunspots with a telescope? _____

_____

5 What is the maximum temperature of the Sun's centre? _____

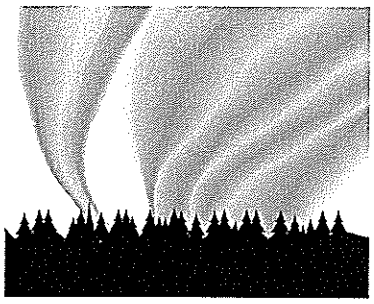
6 What substance is produced in the Sun by the fusion of hydrogen? _____

_____

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The Sun: a massive nuclear reaction II

Northern and Southern Lights



Near the North Pole and the South Pole, spectacular light shows in green, yellow, red and shades of blue and purple, can be seen. The 'Northern Lights' are called *Aurora Borealis*, while the 'Southern Lights' are called *Aurora Australis*. The Sun throws off gas particles that are charged with electricity. When these particles react with gases in the atmosphere and the magnetic poles, the particles glow. The colour produced depends on the gas molecules involved and their distance from the Earth's surface. One of the best places in the world to witness the dancing colours is Canada's Hudson Bay region where the Northern Lights can appear several times a week. Scientists think that one cause of aurorae is sunspot activity. Sunspots are darker, cooler areas on the Sun's surface that contain strong magnetic fields.

Van Allen radiation belts

Around the Earth's equatorial region two 'belts' of charged particles exist. First discovered in 1958, by a scientist named James Van Allen, the belts are made of charged particles that are trapped by the Earth's magnetic field. The particles in the upper belt, found at about 20 000 km above the Earth, come from *solar winds*. The lower belt, at 2000–5000 km above the equator, is made of particles caught up from the Earth's atmosphere. These belts of ionising radiation can damage organisms and interfere with electronic equipment. Space missions are, understandably, designed to minimise the amount of time spent in these belts.

Solar energy

Nuclear fusion reactions in the Sun produce the radiant energy it gives off.

★ **Nuclear fusion** is the combination of two nuclei joining together to form a larger one ('Fusion' means to fuse or join.) In the process huge amounts of energy are released. ★

In stars, hydrogen nuclei fuse at high temperatures to form a helium nucleus and release high-energy electromagnetic radiation. Scientists on Earth have tried to replicate this process without success. Life would not exist on Earth without solar energy. About a third of the energy from the Sun that reaches the Earth's atmosphere is used to power the hydrological (water) cycle. Photosynthesis uses the Sun's energy to convert carbon dioxide and water into glucose and oxygen.

1 In which hemispheres are the *Aurora Borealis* and *Aurora Australis* respectively? _____

2 State one cause of aurorae. _____

3 Where are *Van Allen* belts located? _____

4 What is the composition of Van Allen belts? _____

5 Why would scientists planning a space mission try to ensure a spacecraft spent the minimum time going through the Van Allen belts? _____

6 State two natural processes that utilise some of the Sun's energy. _____ _____

7 Define 'nuclear fusion.' _____

8 Which element is turned to helium in the Sun as a result of nuclear fusion reactions? _____

Research

9 Define 'nuclear fission' and give an example. _____

10 Why might scientists try to carry out a fusion reaction on Earth? _____

11 Find out the names and relative amounts of elements present in the Sun. _____

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The Sun and Fusion

NUCLEAR REACTIONS

The Sun provides energy for life to exist on Earth. It does this by radiating energy generated by a series of nuclear reactions. Nuclear reactions involve the nuclei (centres) of atoms splitting (fission) or joining together (fusion) to form different elements. In both of these processes, mass is changed to energy. Einstein came up with the following famous equation:

$$E = mc^2$$

where E = energy (joules, J), m = mass (kilograms, kg), and c = the speed of light (3×10^6 m/s)

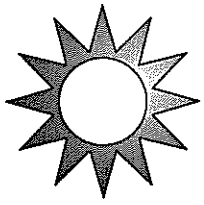
1 Calculate the energy released if 1 gram of matter was completely converted to energy. The Sun gives off 3.8×10^{26} W per second (where W = watts, and watts are joules per second). This is a phenomenal amount of energy. _____

2 Does all of the energy given off by the Sun reach the Earth? Explain your answer. _____

_____

Each atom is itself made up of smaller particles: protons, neutrons and electrons.

	Mass (amu)	Electrical charge	Location
Proton	1	+1	nucleus
Neutron	1	neutral	nucleus
Electron	$\frac{1}{1800}$	−1	orbiting nucleus



The simplest atom is that of the element hydrogen and is shown as ${}_1^1H$ This element has one proton and one electron (it does not have any neutrons). The next simplest element is helium, ${}_2^4He$, and this has two protons, two neutrons and two electrons.

In the core of the Sun, temperatures are so high (millions of degrees) that the elements present exist as nuclei (the electrons have been lost from the atoms).

3 What is the electrical charge on an electron? _____

4 What is the electrical charge on a proton? _____

The initial nuclear reaction in a star is the fusion of four hydrogen nuclei (protons) to form a helium nucleus and particles called positrons (positive electrons).

$$4\ {}_1^1H \rightarrow {}_2^4He + 2\ {}_1^0e$$

The energy released is a counter to the force of gravity pulling the materials inward. Our Sun is estimated to fuse 400 million tonnes of hydrogen to helium every second, but is still expected to last another 5 billion years. The Sun is currently made up of about 75% hydrogen and 5% helium.

As time goes on, other nuclear fusion reactions can occur in a star's core to form heavier nuclei, for example helium nuclei fusing to form beryllium nuclei.

$$2\ {}_2^4He + {}_2^4He \rightarrow {}_4^8Be$$

Another reaction that can take place in the core of a star is the fusion of a beryllium nucleus with a helium nucleus to form a carbon nucleus, ${}_6^{12}C$.

5 Write a balanced equation for the reaction of a beryllium nucleus with a helium nucleus.

_____

6 What is the electrical charge on a neutron? _____

7 State two similarities between nuclear fission and nuclear fusion. _____

8 State two differences between nuclear fission and nuclear fusion. _____

9 If our Sun is half way through its life, what is its estimated total life span? _____

10 What force pulls materials in a star inward? _____

11 Name the simplest element. _____

12 How many protons are there in an atom of hydrogen? _____

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Sunspots

A **sunspot** is any dark, cooler spot that appears from time to time on the Sun's surface.

Sunspots are usually only visible through a telescope but were first recorded by the Chinese thousands of years ago. Galileo Galilei was one of the first people to study the Sun (around 1611) using the then newly discovered *'optick tube'* (unknowingly risking blindness in the process).

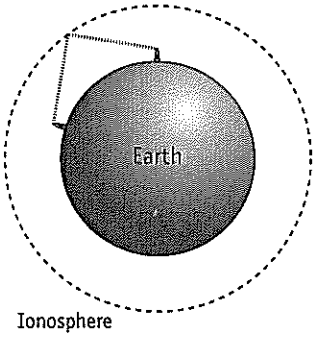
1 What is another name for an *optick tube*? _____

Little research was done on sunspots until Samuel Heinrich Schwabe (1789–1875), a German pharmacist and amateur astronomer, discovered the periodicity (regularly occurring cycles) of sunspots. Schwabe was originally looking for an unknown planet passing in front of the Sun, which he reasoned would be detected as a dark spot. He made observations on every day that he could from 1826 to 1843 and noticed a regular variation (there appeared to be a maximum of sunspot activity about every 11 years). This 'solar cycle' of increasing and decreasing sunspots has been confirmed by many astronomers as ranging from 9.5 to 11 years (averaging 10.8 years) between periods of maximum activity. The Sun's magnetic field reverses its polarity over the same time scale.

Many people have tried to find a connection between the pattern of sunspots and the Earth's weather. Although there doesn't seem to be a connection between sunspot activity and day-to-day weather conditions on Earth, there may be a link with long-term climate patterns. Between 1645 and 1715 there were very few sunspots. This phenomenon (referred to as the 'Maunder Minimum') coincided with very cold conditions in Europe and North America known as the 'Little Ice Age'.

As they wax and wane, sunspots can vary in number from as many as 200 to none. Sunspots can differ in size, with diameters from 2500 km to 50 000 km. Smaller sunspots last a few days and large ones can last for months. Sunspots are cooler (about 4300 K) than the Sun's surface (6000 K). Sunspots are formed when a change in a magnetic field rises to the Sun's surface. The magnetic fields of the Sun can be over two thousand times stronger than those of Earth.

The *ionosphere* is a layer of the Earth's atmosphere lying between 60 km and over 600 km from the surface. It contains ionised (electrically charged) gases that can act as a 'mirror' for some radio waves. The ionosphere can reflect shortwave radio signals from one part of the Earth's surface to another, reaching places farther from the transmitter than might be expected. Sunspots influence the ionosphere. During *solar maxima* the ionosphere is denser than during *solar minima* and communication depending on certain radio frequencies can be affected. Radio astronomers find it difficult to detect cosmic radio signals during solar maxima because the more dense ionosphere reflects incoming signals.



Ionosphere

2 Why should you avoid looking at the Sun directly through a telescope? _____

3 Give a description of a sunspot. _____

4 What is the 'solar cycle' discovered by Schwabe? _____

5 What causes sunspots? _____

6 What was the 'Maunder Minimum'? _____

7 What is the average temperature of the Sun's surface? _____

8 What is the ionosphere? _____

9 What is the best time for a radio astronomer to make observations: during a time when sunspots are many or when they are few? Explain your answer. _____

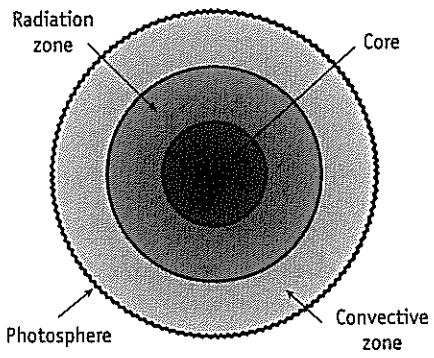
10 What does 'wax and wane,' mean? _____

11 'The ionosphere is an electrically charged layer of Earth's atmosphere.' True or false? _____

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The Sun and Solar Wind

Our Sun ('Sol') is an average-sized main sequence star of G2 stellar class. The Sun is thought to consist of the following layers:



- ☆ **core**: a very dense area where thermonuclear reactions occur and hydrogen nuclei fuse to form helium nuclei. Temperatures in the core may reach 1.6×10^7 K
- ☆ **radiation zone**: where heat is carried from the core outwards by radiation.
- ☆ **convective zone**: where heat is transferred by convection of gases.
- ☆ **photosphere**: the luminous surface of the Sun. Less than 5000 km deep with temperatures ranging from approximately 4700 K to 6000 K. Sunspots occur on the photosphere.
- ☆ **chromosphere**: a layer extending about 130 000 km above the surface. Pressure and density decrease with height while the temperature rises to 50 000 K.
- ☆ **corona**: this is diffuse, invisible under normal conditions (it can be studied during eclipses) and extends for several million kilometres. It emits high-energy particles ('solar wind').

The solar wind leaves the Sun at speeds of up to 600 km/s. It spirals out with the rotation of the Sun, affecting the electrically-charged tails of comets, and curves around the magnetic fields of Earth, and around Jupiter. (These are the only planets known to have radiation belts associated with strong magnetic fields).

Sometimes a small area near a sunspot can suddenly explode, giving off gamma rays, X-rays, radio waves and electrically charged particles in an event called a *solar flare*. The particles can reach Earth and cause disruption to electronic communications. They are thought to be responsible for the *Aurora Borealis* and *Aurora Australis* (Northern Lights and Southern Lights) and even cause the outer layers of the Earth's atmosphere to expand. Due to the potential danger to satellites and space missions, organisations that send objects, or astronauts, into space need to check 'space weather'. Sunspot and solar flare activity is monitored and reported by many observatories. One observatory in Australia (Learmonth, Exmouth, WA) studies solar activity.

★ **Helium** is named after Helios, the sun god of Greek mythology. ★

1 What is the 'solar wind'? _____

2 If the Earth is 150 million km away from the Sun and the solar wind travels at a mean speed of 400 km/s, how long will it take to reach the Earth? Show your working. _____

_____

3 Draw a labelled diagram of the layers of the Sun, showing the location of the core, radiation zone, convective zone, photosphere and chromosphere.

4 What is the maximum temperature in the core of the Sun? _____

5 When can the 'corona' of the Sun be seen? _____

6 Name three things given off by a solar flare. _____

7 Name two effects that solar flares have on the Earth. _____

8 In which region of the Sun do sunspots occur? _____

9 A telecommunications company plans to send a satellite into orbit. Why might the company need to take account of solar activity? _____

10 What is another name for our Sun? _____

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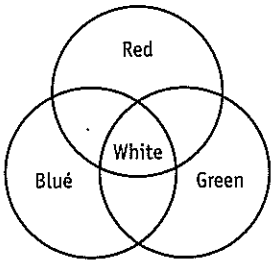
Spectra

A prism is a block of transparent material that can refract (bend) light. If a beam of white (visible) light is shone through a triangular prism it can be split into a spectrum of colours, such as we see in a rainbow.

1 Name the seven main colours of the spectrum. (*Hint:* ROYGBIV) _____

A **spectrum** is the pattern (or bands) of colours produced when light from a particular source is split into its components.

Experiment: How to create white light. You will need three torches (or ray-boxes) and three pieces of cellophane (one red, one blue and one green), large enough to cover the glass head of each torch. Wrap a piece of red cellophane over the glass of one torch, blue over another torch and green over the last torch. When you turn on your torches, they will now produce red light, blue light and green light. In a darkened room, direct the coloured spots from your torches toward a wall, bringing them closer together until you form a pattern resembling a 3-leafed clover. At the centre of the shape, where all three colours combine, you will see white light.

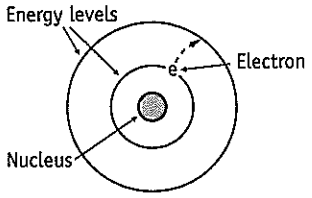


2 Would you get the same result if you mixed paints of the same colour?

_____

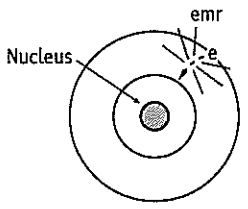
The split, white light that comes out through a prism appears in clearly defined bands. Depending on the source of the light, the banding patterns (spectra) can all be different (though the colours are always in the same order). Scientists can tell a great deal from the spectra given off by stars. The more alike the spectra are, the more alike the stars are. Bands reveal:

- ☆ how far away a star is
- ☆ how hot a star is
- ☆ what a star is made of
- ☆ how old a star is
- ☆ how fast a star is moving (or if it moves at all).



a) Electron jumps to next energy level for a time.

A knowledge of atomic theory can help us understand why the spectra of stars are different. If an atom has energy put into it by being heated, electrons become 'excited' and move into higher energy levels. These electrons fall back into lower, more stable, energy levels and, in the process, emit electromagnetic radiation (emr) of a specific frequency. Each element has its own individual spectrum. Before the element helium was discovered on Earth, its presence was discovered in the Sun by studying the Sun's spectrum. The source of energy in stars is the fusion of hydrogen atoms into helium atoms.



b) Electron falls back to its original level, emitting emr as it falls back.

3 What is a prism? _____

4 Name four things astronomers can tell about a star by studying its spectrum. _____

_____

5 Which three colours combine to give white light? _____

6 What does 'emr' stand for? _____

7 Whereabouts in an atom are electrons located? _____

8 What is an 'excited' electron? _____

9 What are the two most common elements making up a star? _____

10 'Hotter stars have different colours compared to cooler stars.' True or false? _____

11 'Hydrogen atoms split to form helium atoms in the nucleus of a sun.' True or false? _____

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Stars I



About 5000 stars can be seen with the naked eye and the longer we stay out watching the night sky, the more stars we will see. We see stars because their light results from massive nuclear reactions occurring within them. These reactions involve the fusion (fusing together) of hydrogen atoms into helium atoms. When the reaction occurs, and helium is formed, heat and light are also created.

1 As well as heat and light, what forms of electromagnetic radiation are emitted during nuclear fusion?

_____

The brightest star in our galaxy is *Sirius*. *Sirius* is 26 times brighter than our Sun and is 8.7 light years away (which is just as well or we would all be toast!). Groups of stars forming recognisable patterns are called *constellations*. In the past, constellations were named by ancient Greek and Babylonian stargazers using a 'joining the dots' technique. Many of the shapes were imagined to resemble creatures from ancient mythology and were named accordingly.

2 The best-known constellations are those in the *Zodiac*. Name the 12 *signs of the Zodiac*.

_____

Among astronomers, one of the most well known constellations is the spectacular *Pleiades* cluster located just over 400 million light years from our Solar System. The Pleiades is 50-60 million years old (a relative youngster!) and much of the gas from which it was formed still surrounds it. Although Pleiades' hundreds of stars have been seen through the telescope, only seven can be seen with the naked eye. These are affectionately called *The Seven Sisters*. Stars are still being formed in the Orion (the Hunter) constellation. Orion contains a massive red star called *Betelgeuse*. *Betelgeuse* (*alpha Orionis*) is about 500 times bigger than our Sun. Though it is 310 light years away, it is the tenth brightest star in our skies!

3 Give your own definition of a constellation. _____

Brighter stars are usually closer to us, but this is not always the case. Clouds of 'space dust' can often block out a star's light making it appear dimmer. It is easier to see stars at night, and the less light there is around you, the better you will see them. Stars continue to shine during the day, but the brightness of the Sun makes them too difficult to see.

SHOOTING STARS

It is not unusual to see what looks like a star shooting across the sky. As suddenly as the star appears, it vanishes. Shooting stars are *meteors*. A meteor is a piece of space rock moving at such a rapid speed that it heats to a burning hot temperature. Usually the meteor is incinerated in the sky and the tiny remaining dust particles disperse and are lost. Sometimes a meteor is either very large or does not completely burn up. In this case, a *meteorite* falls, occasionally landing on Earth. If the meteorite is particularly large, it can hit the ground at such a speed it creates an enormous crater. *Asteroids* are large fragments of rock scattered between Mars and Jupiter. Some can be up to 1000 km in diameter.

4 Where would you see the most stars, in the city or in the country? Explain why. _____

5 Draw the constellation found on the Australian and New Zealand flags.
Name it and join the dots to show how it came to be named.

6 What is another name for the 'Seven Sisters'? _____

7 Which is the brightest star in our galaxy? _____

8 Explain the difference between a *meteor* and a *meteorite*. _____



Research

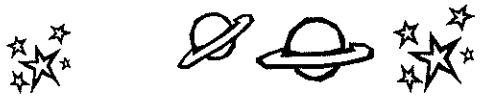
9 The colour of a star can be some indication of its age. Would a cluster of bright blue/white stars be young or old? Explain your answer.

10 One hypothesis exists that dinosaurs (and other life) died out about 60 million years ago due to either a large asteroid or huge meteor showers hitting Earth. Write a short report on this theory.

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Stars II

To say that a star 'twinkles' is a gross understatement. The words 'ignite', 'explode', 'blaze', 'crash', and 'erupt' present a more accurate description than 'twinkle'. Starlight is produced by spectacular nuclear reactions. These fusion reactions compress hydrogen atoms to produce helium, and energy in the form of heat and light. It is when these explosions run out of fuel, and changes take place, that a star is said to die. Stars don't really die; they change from one form (the star) to others, depending on the manners in which they ended their lives as stars. But first, a star must be born ...



THE BIRTH AND DEATH OF STARS

There are several common phases, or a *main sequence* of events, in the life of a star. Stars are born amid a cloud of contracting gases. A gas cloud forms a *plasma* which contains electrons and nuclei at temperatures greater than 10 000°C. It can be compared to water condensing on a ceiling when the kettle is boiling. The more the hydrogen atoms compress, the greater the frequency of them bumping into each other. This bumping together leads to many *fusion* reactions.

Hydrogen atoms fuse (join) to form helium atoms. Along with helium, heat and light are given off. With increasing heat (tens of millions of degrees), continuing contraction and hugely increased pressure, more reactions occur. When enough fusion reactions have occurred, the heat and rate of reactions become so rapid that a runaway reaction occurs and the star explodes into life.

$$4\ {}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2\ {}^{0}_{1}e$$

$${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}N$$

One of the reactions occurring in a star is the fusion of hydrogen nuclei (protons) to form helium nuclei and positrons (positive electrons). Isotopes are atoms of the same element having different numbers of neutrons. Another typical stellar fusion reaction involves atoms of deuterium (isotopes of hydrogen) forming helium, neutrons and energy. Millions of megajoules of energy are released when just a few grams of hydrogen fuse.

1 A joule (J) is a unit of energy. How many joules are in a megajoule? _____

2 What does the term 'stellar' describe? _____

Fusion energy can be compared to the burst of energy that makes a group of children want to run off in all directions when entering an amusement park. The star's gravity is like the parents who continually pull the children back together into a manageable group. The outward movement of the fusion energy released is roughly balanced by the inward pull of the star's gravity, holding the unit together.

The stronger the gravitational pull, the more densely packed the molecules (mainly hydrogen) become. The greater the density, the greater the frequency of fusion reactions (producing heat and light as a by-product). Eventually 'the whole thing goes up!' and a star is born.

As long as there is hydrogen to 'burn' a star will live. The more gas (fuel) there is, the greater the rate of reaction, producing starlight of a greater intensity. After the star has exploded into life, for a time it follows a predictable 'main sequence' of events and is, therefore, called a *main sequence* star.

3 Which element is the starting point in fusion reactions in stars? _____

4 What is the symbol for the element helium? _____

5 What is an 'isotope'? _____

6 Scientists have been unable to carry out the same fusion reactions that occur in the Sun. Give a reason for this. _____

7 Name the force of attraction that all bodies have for each other. _____

8 Does hydrogen actually burn in a star? Explain your answer. _____

Research

9 Plasma has been described as the fourth state of matter. Write a brief account of plasma. (*Note:* do not confuse this plasma with the 'other' plasma that is a constituent of blood.)

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Stars III

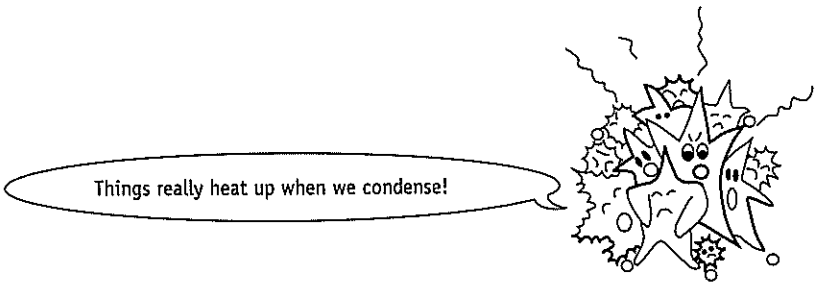
1 Use the words listed to fill in the spaces in the following passages.

alone	clouds	clusters	constellations	cross	destroy
helium	hydrogen	large	main	nebula	nova
nuclear	particles	plasma	southern	star's	temperature

WHERE DO STARS COME FROM?

Stars are formed from *nebulae* which are giant **a** _____ of helium gas and other **b** _____. The most well known **c** _____ which can be seen with the naked eye is near Orion. When the clouds contract forming a heavy, more compact, cloud and the **d** _____ increases, the particles come together causing a **e** _____ reaction to occur. **f** _____ is a collection of charged particles that resembles a gas. Stars are seldom born **g** _____. Usually these massive reactions create many **h** _____ of stars. The brightest clusters can be seen in the **i** _____ hemisphere. The next time you are looking at the night sky, be patient and you should eventually see a thin 'cloud' that is actually a cluster of stars! Clusters of stars that travel together are called **j** _____. One of the brightest constellations in the southern hemisphere is called Alpha Centauri. The Southern **k** _____ is found near Alpha Centauri.

THE LIFE OF A STAR



Once it is born, a **l** _____ life is predictable. When it stops condensing the star begins to burn at a steady rate, which depends on what it is made of. At this stage it is called a **m** _____ sequence star. Main sequence stars grow brighter and brighter, all the time undergoing the nuclear reaction of converting hydrogen to energy and **n** _____. Sometimes explosions occur that are more intense than usual and these explosions can **o** _____ a star. If not, the star burns even more brightly and is called a **p** _____. Eventually all the **q** _____ is burned and the core of the star begins to shrink. The core is still surrounded by heat and this heat is pulled inward. The outer part of the star cools and spreads out. In the same way as a bush fire expands to find more fuel, stars expand looking for more hydrogen to burn. The star becomes a **r** _____ cloud with a hot core and is called a *red giant*.

2 'If a celestial object can be seen with the naked eye it can not be seen without a telescope.' True or false? _____

3 'The nuclear fusion process occurring in stars is easily duplicated on Earth.' True or false? _____

4 'Helium and energy are products of the fusion of hydrogen.' True or false? _____

5 'The life cycle of a main sequence star follows a predictable pattern.' True or false? _____

6 'A 'shooting star' is a star that has just exploded.' True or false? _____

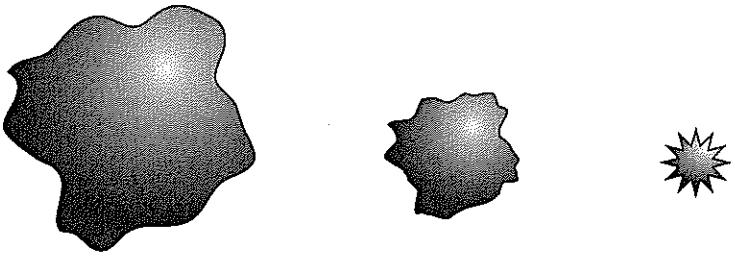
7 'All stars are of equal brightness.' True or false? _____

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The Life of a Star

Because stars may have existed for billions of years, Astronomers have been found to theorise about their formation. It is thought that stars arise, over millions of years, as huge clouds of hydrogen (H) gas and dust which condense due to the force of gravity. In these clouds, known as *nebulae,* as hydrogen atoms squeeze together, the temperature of the cloud rises to approximately 1 000 000°C and fusion reactions start to occur. Fusion reactions involve the nuclei of atoms joining (fusing) to form larger nuclei. The fusion of hydrogen (H) produces helium (He), heat and light energy. A great 'tug-of-war' occurs within the protostar as the heat causes it to expand and gravity pulls it inwards. Gravity is, initially, the stronger of the two forces so the overall direction of pull is inward.

The size of the nebula is critical in the formation of a star. If the mass is less than 0.1% of that of our Sun, not enough pressure will be present to start the fusion reaction. If the mass of the nebula is greater than 60 solar masses then the star is likely to be very short-lived.



Nebula Contracting nebula Star

As long as there is enough hydrogen present to use as a fuel, the tug-of-war continues and, the more fuel there is, the brighter the star appears. Stability occurs when the star eventually is radiating the same amount of energy as is generated from its core. Once they have exploded into life, 90% of stars follow a predictable sequence of events and are therefore known as *main sequence* stars. Our Sun is a main sequence star. Main sequence stars burn steadily for as long as their fuel supply is available. However, not all stars make use of their fuel at the same rate. Stars change over their lifetime. Small stars use their fuel more slowly than large ones and, hence, change more slowly as well. The size of a star, as well as its brightness and surface temperature, allow us to make some predictions about its eventual fate.

1 Is gravity an attractive force or a repulsive force? Explain your answer. _____

_____

2 What is a nebula? _____

3 What does this definition describe: 'The union of light nuclei to form heavier nuclei, releasing vast quantities of energy'? _____

4 What is the symbol for the element hydrogen? _____

5 What is the symbol for the element helium? _____

6 What is a 'protostar'? _____

7 What is meant by a 'main sequence' star? _____

_____

8 What is the main fuel in stars? _____

9 When does a forming star become stable? _____

10 Which use up fuel more quickly, small stars or large stars? _____

11 What characteristics of a star help predict how it will behave? _____

_____

12 Does heat cause a star to contract or expand? _____

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The Fate of a Star

At some point a star's fuel runs out and, without it, fusion reactions will cease. Gravity continues to pull everything inwards, producing a dense mass. At this point many new, heavier, elements are forming, such as neon (Ne), iron (Fe), oxygen (O) and silicon (Si). The star does not give up without a fight. As the main sequence of events draws to a close, and the remaining fuel decreases, stars of average size swell, spreading out as far as possible in an effort to reach whatever fuel is nearby. If the star is between 1.4 times and 3 times the size of our Sun, when it expands in search of more fuel, an enormous *red giant* is produced. As a result of spreading itself thin in its effort to reach more fuel, red giants have relatively cool surface temperatures. The red giant cannot expand continually. When all fuel is used up, the remaining white-hot 'embers' can collapse into a *white dwarf*. White dwarfs are difficult to observe because they have a very low luminosity. Over time, the white dwarf cools, producing a cold, dead *black dwarf*.

1 What is meant by 'very low luminosity'? _____

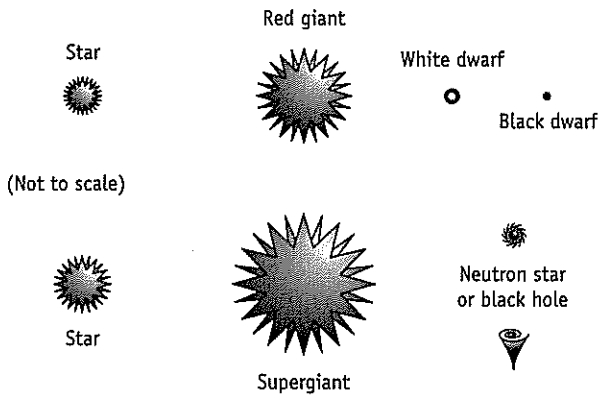
2 Why do you think that astronomers use terms like 'dwarf' and 'giant' to describe stars? _____

_____

As fuel is exhausted by larger stars (those greater than three times the mass of our Sun) those stars follow a different sequence of events after the red giant phase. An even larger *supergiant* swells to life and can (depending on size) form a *neutron star* or, in some circumstances, a *black hole*.

A neutron star is a very dense, rotating, object that emits electromagnetic radiation and is called a *pulsar*.

A black hole is a body with an immensely strong gravitational field.



3 How can astronomers observe neutron stars? _____

4 Copy the diagram above of the sequence of how stars change.

5 Why isn't the diagram drawn to scale? _____

6 State one way that astronomers could differentiate a pulsar from a 'normal' star. _____

7 What is the major fuel used by stars in the process of fusion? _____

8 What is the chemical symbol for iron? _____

9 What is the chemical symbol for oxygen? _____

10 What is the chemical symbol for neon? _____

11 What is the chemical symbol for silicon? _____

12 What is the force that pulls the materials of a star inwards? _____

13 Arrange in the correct order: main sequence star → black dwarf → white dwarf → red giant

_____

14 'Our sun is a "main sequence" star.' True or false? _____

15 'Red giants are larger than supergiants.' True or false? _____

16 'Black holes are areas where gravity doesn't exist.' True or false? _____

17 'The lifetime of a main sequence star is measured in thousands of years.' True or false? _____

18 'Black dwarfs are stars that emit little or no radiation.' True or false? _____

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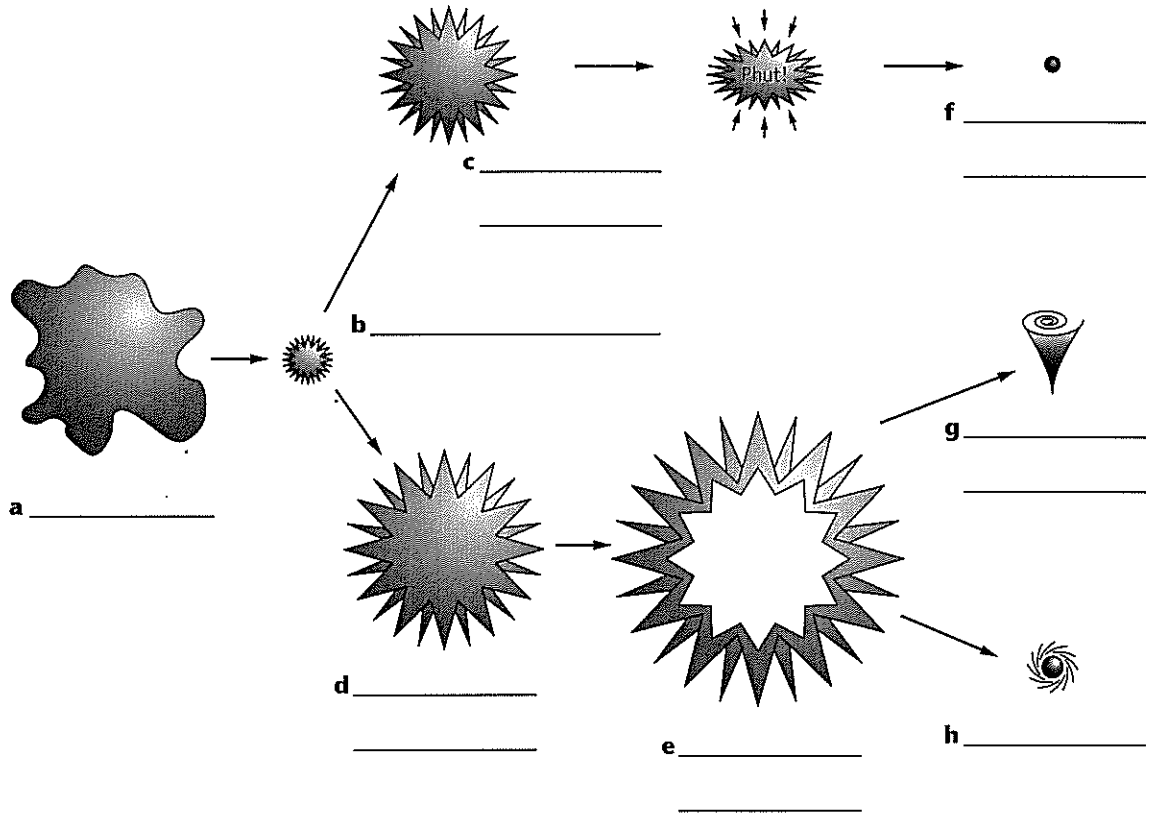
Red Giants: giants, dwarfs and black holes

Depending on the size of a star, and what it is made of, a red giant can burn for millions of years. Eventually it collapses, throwing pieces of white material everywhere. The pieces are lost in space, each becoming a tiny *white dwarf*. Although white dwarfs are small compared to red giants, they have a lot of mass in a small volume (that is, they are very dense). In the same way that shearers can pack enormous amounts of wool into fairly small bundles, so does a red giant 'pack a lot of star' into a white dwarf. If a white dwarf were the size of an apple, it would weigh more than a large herd of elephants!

If the red giant star is heavier than usual, at least three times heavier than our Sun, it reaches an enormous *supergiant* stage. Like any red giant, the heavy star expands to an incredible size in a final search for more fuel. Eventually it undergoes a massive explosion called a *supernova*. Supernovae are so intense that they can be one billion times brighter than our Sun and can create the mysterious *black holes* and *neutron stars*. It is at the point just before the supernova explosion occurs that the awesome *supergiant* is created. From this point, one of the following two paths can be followed.

i If the star was *twelve times heavier than the Sun* (or more) and the supernova explosion did not destroy it, a super-compressed neutron star can result.
ii If the star was destroyed, a black hole ('collapsar') is created.

1 Below is an illustration of the birth and death of a star. Copy the diagram and label the steps.



2 What does this define: 'A hypothetical invisible area having a small volume but an immense gravitational field'?

_____

3 What does this define: 'The explosion of a very large star releasing large amounts of energy'

_____

4 What does this define: 'A bright star, much larger than the Sun, with a relatively low surface temperature?'

_____

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Black Holes

It takes a big star to become a black hole! The death of a supergiant produces a massive gravitational force so powerful that the star 'sucks itself up' creating a hole in space. This hole continues to have such a strong inward pull that even light cannot escape, giving the hole in space the name *black hole*. Black holes, in a generally dark universe, are invisible to the eye but they give off immense amounts of heat energy and radio waves, revealing their presence. Like an invisible magnet, black holes pull not only themselves, but also anything nearby, into their depths. Even light is trapped because it would need to travel faster than it already does in order to escape. Since light travels faster than anything else we know of, all other matter passing by the black hole must also be swallowed up.

1 Light cannot escape a black hole. What does this say about the rate at which gravity is pulling inward?

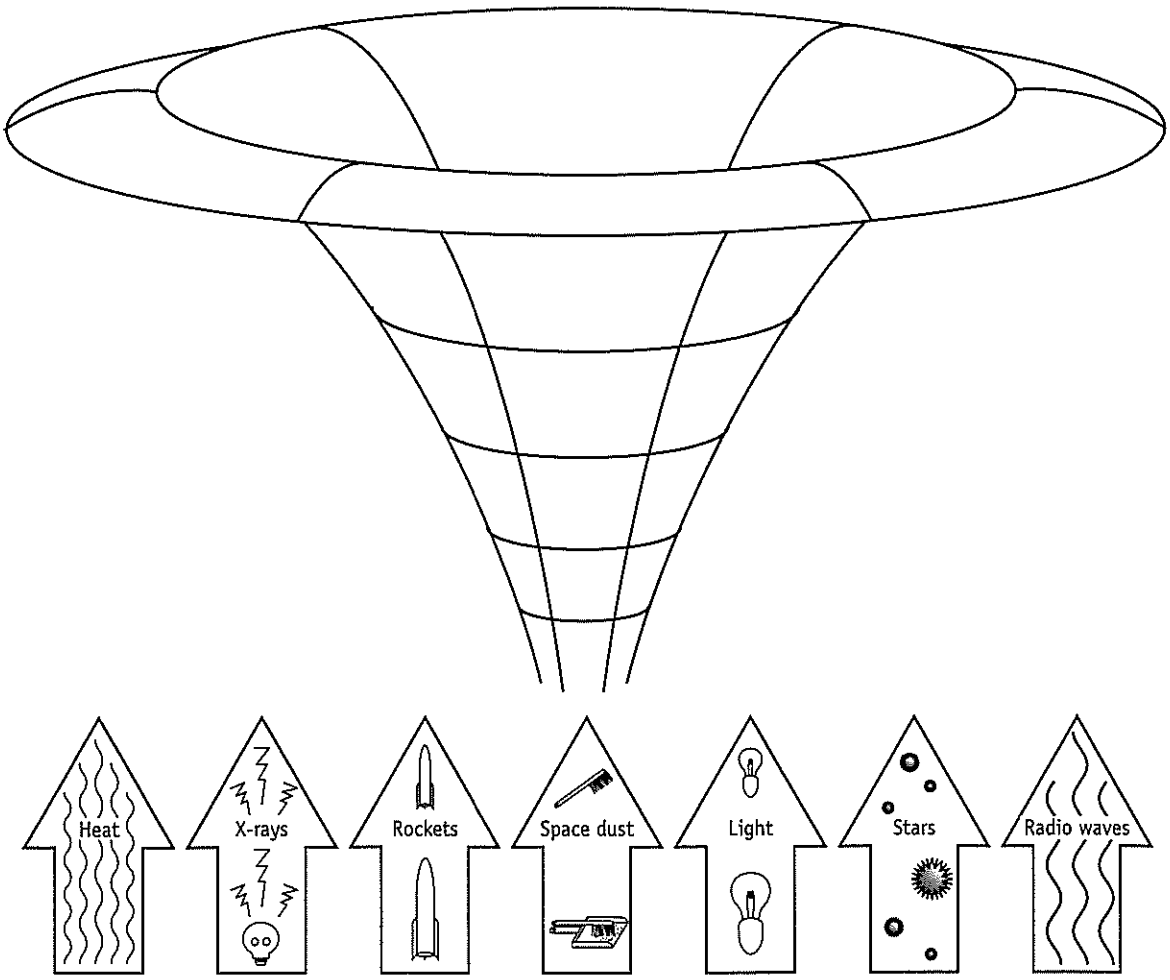
_____

EXERCISE: BLACK HOLE MODEL

The motion of an object just before it is swallowed up in a black hole can be illustrated using the kitchen sink. Put the plug in the plug hole and fill the sink with water. Sprinkle a *few* pieces of confetti over the surface of the water.* As the confetti approaches the drain, it will swirl before it disappears down the hole. Stars and other matter behave the same way when approaching black holes.

*Only use a few pieces of confetti or you will need to ring a plumber!

2 **Beware the black hole!** Cut out the arrows below and paste them in the correct direction on the diagram according to how the black hole would affect them.

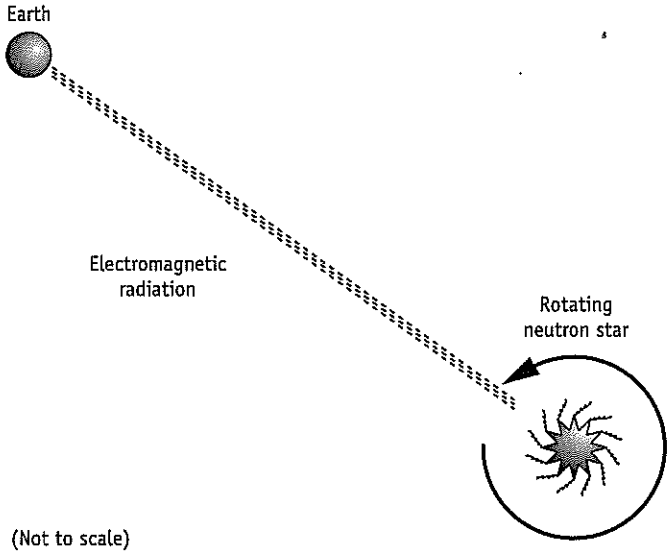


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Neutron Stars and Pulsars

Pulsars have been called the lighthouses of deep space. They are rotating, dense, exploded neutron stars that emit huge amounts of electromagnetic radiation, including light, X-rays and radio waves. The explosion is so fast, and so intense, it sets the remains of the star, which are mostly neutrons, spinning up to 1000 times per second. Since there is no force (such as gravity) acting to stop it, the star just continues spinning. The light pulses at a steady rate and the pulses remain constant over time.

In the same way as the bulb in a lighthouse revolves, creating a flash of light, so the neutron star spins. A famous pulsar is the Crab nebula. In the year 1054 a supernova explosion occurred that was so powerful that stargazers in China, the Middle East and North America witnessed it. The pulsar was visible night and day for nearly one month even though it occurred 6500 light years from Earth! The Crab still pulses at about 30 flashes per second and is visible as a misty patch in the skies.



EXERCISE: DEMONSTRATE THE MOTION OF A PULSAR

In a darkened room, ask another student to turn at a steady rate while holding a lit torch. The light will 'pulse' as the student faces the class, and dim as it faces the opposite direction. Now imagine the student with the torch spinning at a rate of up to 1000 times per second.

1 Define 'nebula'. _____

2 Define 'comet'. _____

3 Define 'asteroid'. _____

4 Define 'meteor'. _____

5 Define 'black hole'. _____

6 Define 'neutron star'. _____

7 Define 'red giant'. _____

8 Define 'supergiant'. _____

9 Define 'white dwarf'. _____

10 Define 'red dwarf'. _____

11 Define 'black dwarf'. _____

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Star Classification

The first people to combine a camera with a spectroscope were the father and son team of John and Henry Draper in the 1870s. Their work was carried on by Edward C. Pickering who, by 1918, had listed the spectra of over 200 000 stars. Using details about luminosity and composition, stars are classified into several common groups, or classes, depending on their colour (*spectral types*). Each difference in colour is directly related to the *surface* temperature of the star. The hottest stars are blue-white, with surface temperatures of greater than 50 000°C. The cooler the star, the more red it appears.

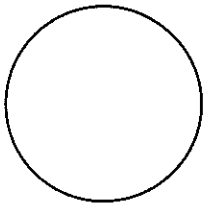
The Pickering empirical classification has become the accepted way to describe stars:

- ☆ M orange/red (coolest)
- ☆ K orange
- ☆ F/G yellowish
- ☆ A white/bluish-white
- ☆ O/B white (hottest)



The temperatures (and colours) of stars are a gradual progression and the divisions between groups are arbitrary. To remember the correct sequence, from hottest to coolest (O B A F G K M), think of the following catchphrase: '***O**nly **B**old **A**nd **F**earless **G**iants **K** **M**artians*'. Further subdivisions rank these classes from 0–9.

Other schemes include stars described as X, W, N, R and S, although these are very rare and represent only a fraction of a per cent of the total number of stars so far observed and classified.



1 Of the stars that have been classified by their spectra, B stars represent about 3%, A stars are 27%, F are 10%, G are 16%, K are 37% and M are 7%. Draw a sector graph (pie chart) to show these figures.

2 Why weren't Type O stars mentioned in Question 1?

_____

3 Use your knowledge about spectral types to match the jumbled surface temperatures in the box on the right with the correct star.

	Star	Spectral type	Temperature (K)	Distance (light years)
a	Sirius	A0		9
b	The Sun	G2		
c	Betelgeuse	M2		520
d	Spica	B1		220

3000
25 000
6000
11 000

4 What else can we predict about *Sirius* from the information above? _____

5 What else can we predict about *Betelgeuse* from the information above? _____

6 What else can we predict about *Spica* from the information above? _____



Research

7 In which constellation is *Sirius* located? _____

8 In which constellation is *Betelgeuse* located? _____

9 In which constellation *Spica* located? _____

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Hertzsprung–Russell: (H–R) diagram

In 1905, two astronomers, named Ejnar Hertzsprung and Henry Norris Russell, photographed the spectra of light produced by many stars. Hertzsprung (a Dane) studied the magnitude and colour of stars in the Hyades cluster (part of the Taurus constellation) because they were all approximately the same distance from Earth.

1 Why might it be an advantage for all the stars studied to be the same distance from Earth?

_____

Russell (an American) studied several stars close to the Earth whose distances were known from their parallax and red shift. The photographs were used to work out how the stars were organised in the skies. Hertzsprung discovered a link between a star's true brightness (luminosity) and colour and, later, between brightness and temperature. Colours ranged from bright blue/white to red. The brightest, bluest stars were revealed to have the shortest star life (10^6 to 10^7 years), and lightweight stars lived longer than heavy stars. Our Sun is believed to have a life of around 10^{10} years.

2 Write 10^6 years in expanded form. _____

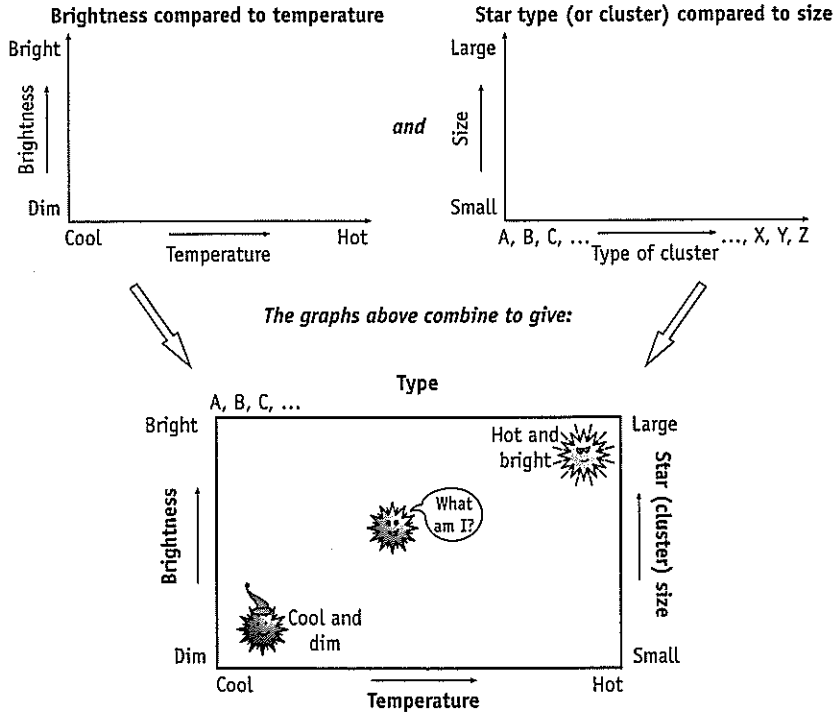
3 Write 10^7 years in expanded form. _____

4 Write 10^{10} years in expanded form. _____

5 Why might bright stars have a shorter life than duller stars? _____

_____

Between 1911–1913 Hertzsprung and Russell independently developed diagrams which illustrated the link between a star's luminosity and its surface temperature. Each man was unaware that the other was doing the same work and reaching the same conclusions. When the data was plotted for each star, patterns occurred and, from this, a star's composition could be predicted. Eventually, the men worked together and their diagrams were so similar that they created a new one, in 1913, calling it the H–R Diagram. When the logarithm of absolute magnitude is plotted against surface temperature (which, relates to spectral type) most stars fall into a broad band called (by Hertzsprung) the 'main sequence' because it is thought to represent the evolutionary path of the vast majority of stars. Some star types don't fit into the 'main sequence' — these tend to be bright, large stars (giants and supergiants) and the extremely dense 'white dwarfs'. The H–R diagram is still used today.

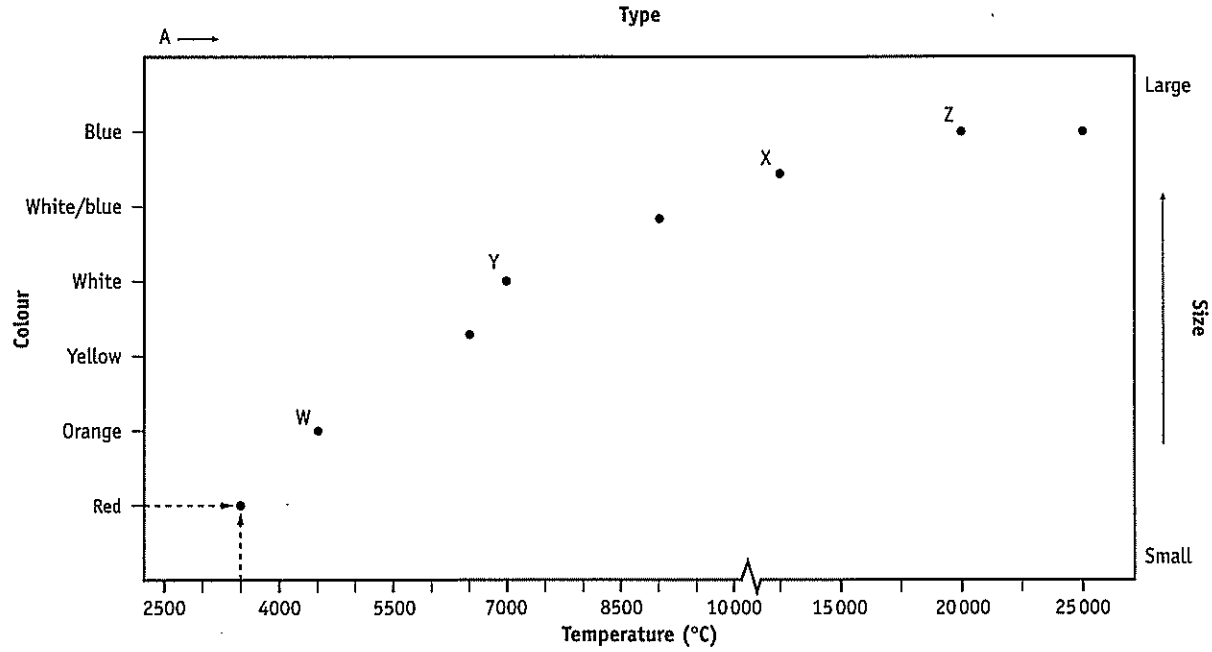


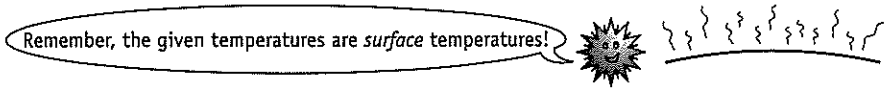
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Hertzsprung–Russell: (H–R) diagram

The horizontal *x-axis* illustrates surface temperature, given as spectral type (sometimes called the *colour index*). The vertical *y-axis* shows luminosity (brightness). By piecing together the information we have about each star, and neighbouring stars, scientists create a picture of our universe that allows us to answer more questions about it.

Below is a graph based on the H-R Diagram. Use the graph to answer the questions below.





6 Stars of which colour are the hottest? _____

7 Stars of which colour are the coldest? _____

8 What colour do you expect a star with a temperature of 30 000°C to be? _____

9 Would you expect to find a blue star with a temperature of 2500°C? _____

10 Would you expect to find a yellow star with a temperature of 5500°C? _____

11 What colour and temperature do you expect each of W, X, Y and Z on the graph above to be? Write your answers in the table below.

Star	Temperature	Colour
W		
X		
Y		
Z		

12 Use your knowledge of our Sun to draw a little sun on the graph in what you would expect to be the correct position.

13 Indicate in blue where you would expect to find main sequence stars on the graph.

14 Indicate in red where you would expect to find red giants on the graph.

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Galaxies

Galaxies are large groups of stars (and other material) that move together in space. Each galaxy identified is given a name and a number. This number is called the 'Messier number', named after the French astronomer Charles Messier (1730–1817). Our galaxy is called the *Milky Way*. Every individual star we see at night is part of the spiral-shaped Milky Way galaxy. Our Solar System is found on the outer edge of the Milky Way.

Astronomers estimate there could be as many as 100 000 million galaxies beyond ours. A galaxy called Andromeda is known by the Messier system as M31. Andromeda can be seen by eye (if you know where to look) but is often mistaken for a nebula (cloud of gas or dust). Andromeda is 2×10^6 light years from our galaxy and is about twice its size.

THE MILKY WAY: THE GALAXY WE CALL HOME

The Milky Way is a typical galaxy. Its name comes from the faintly luminous band it appears as in the night sky, which some say resembles a blurred pattern of flowing milk. The Milky Way is divided into three main parts: the *halo*, the *nucleus* and the *disc*. The halo contains over 10 000 stars, many of them the oldest stars in our galaxy. Most of the stars in the halo are found in clusters, rather than alone. The nucleus, or 'heart', of the Milky Way contains over 5 million stars and is estimated to measure 20 000 × 10 000 light years in size. The youngest stars are found in the disc, which is estimated to have a diameter of 100 000 light years. In the disc drift hydrogen clouds which have yet to form into stars. Where temperatures in the hydrogen clouds reach 10 000°C or more, the cloud glows and is called an emission *nebula*.

1 Draw a labelled diagram of the Milky Way galaxy. Include dimensions and state whether it is a view from the top or from the side.

The term 'nebula' (plural 'nebulae') comes from the Latin word for cloud. In the early days of astronomy, the term was applied to all objects in space that had a diffuse (hazy) appearance.

A constellation is a group of stars that seem to make a pattern. The centre of the Milky Way is in the direction of the constellation of Sagittarius. The 'Great Nebula' is a brightly lit nebula in the constellation of Orion. Some nebulae do not emit light. One of the most famous dark examples is the 'Horsehead' nebula (also in Orion), named for its distinctive shape.

2 What is the shape of the Milky Way? _____

3 What is the name for the galaxy known as M31? _____

4 What is the diameter of the Milky Way? _____

5 Why are light years, rather than kilometres, used to measure astronomical distances?

_____

6 Where are the oldest stars in our galaxy located? _____

7 What makes up a nebula? _____

8 Name two nebulae. _____

9 Name a constellation. _____

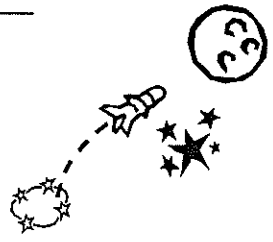
10 'All nebulae are luminous.' True or false? _____

11 'The Milky Way is unlike any other galaxy.' True or false? _____

12 'Our Solar System is at the edge of the Milky Way.' True or false? _____

13 'Galaxies only contain stars.' True or false? _____

14 'The Andromeda galaxy resembles a nebula.' True or false? _____



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